BABU BANARSI DAS UNIVERSITY



PROJECT ON
IBM COGNOS & POWER BI

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Contents

1. Introduction and Objective

2. Dataset Description

3. Data Preparation

4. IBM Cognos Analytics Work

5. Power BI Work

6. Findings and Insights

7. Recommendations

8. Limitations

9. Conclusion

10. References

1. Introduction and Objective

1.1 Business context

RetailPulse is a retail business that sells products in several product categories through stores located in different regions. Like most retailers, its managers need to know which regions and categories drive revenue, how sales change over time, and which customers are most valuable. Raw transaction records cannot answer these questions on their own. They must first be cleaned, summarised and presented in a form that decision makers can read quickly.

1.2 Questions to answer

- Which regions and stores generate the highest and lowest Sales Amount?
- Which product categories sell best, and which are weakest?
- How do sales change month by month and year on year?
- How are customers split between new and returning, and across loyalty segments?
- How can each regional manager see only their own region's figures?

1.3 Objective

The objective of this report is to carry out a descriptive analysis of the RetailPulse data using two tools. IBM Cognos Analytics is used to build list, crosstab, prompt and dashboard reports. Microsoft Power BI is used to prepare the data, build a data model, write DAX measures, apply time intelligence, secure the data by region, and publish an interactive report. The report ends with findings and practical recommendations.

2. Dataset Description

The analysis uses the RetailPulse dataset provided for the course. In Cognos Analytics the data was accessed through the RetailPulse Sales package and the RetailPulse Customers data module. In Power BI it was imported from the workbook RetailPulse_Sales.xlsx, which contains three tables: Sales, Product and Store.

Item	Details
Source	RetailPulse course dataset (Cognos package, data module and Excel workbook)
Tables used	Sales (fact table), Product (dimension), Store (dimension), Customers (Cognos data module)
Number of rows	Sales: [7,13,19,72,955]; Product: [food,clothes,electronic]; Store: [david store]; Customers: [all type
Time period covered	[01-04-2025] to [02-05-2025]

Table 1: Dataset overview

Field	Table	Meaning
Sales Date	Sales	Date on which the sale took place

Field	Table	Meaning
OrderID	Sales	Identifier of an order; used to count distinct orders
Region	Store	Geographic region of the store (for example North, South)
Store Name	Store	Name of the store that made the sale
Product Category	Product	Group to which the product belongs
Quantity	Sales	Number of units sold in the transaction
Sales Amount	Sales	Revenue value of the transaction
ProductID, StoreID	Sales	Keys that link each sale to the Product and Store tables

Table 2: Key fields and their meaning

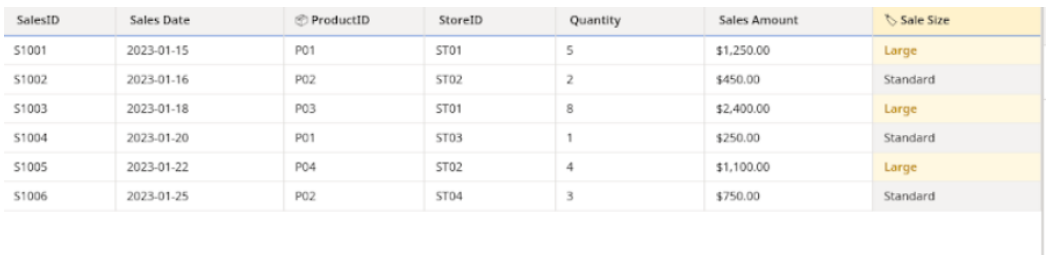
Column names in the workbook may differ slightly from those above; the formulas in this report use the names that appear in my file.

3. Data Preparation

Data preparation was done in the Power BI Power Query Editor. Every step is recorded in the Applied Steps list, so it can be repeated automatically when the data is refreshed. Table 3 lists each step and the reason for it.

Step	Action	Why it was needed
1	Imported Sales, Product and Store tables from the Excel workbook (Home > Get data > Excel workbook)	Brings all source tables into one model using Import mode
2	Set data types: Sales Date as Date, Quantity as Whole Number, Sales Amount as Fixed decimal number	Visuals, sorting and DAX calculations only work correctly when types are right
3	Removed blank rows and duplicate rows from Sales	Blank rows add no information; duplicates would overstate sales
4	Replaced null values in Quantity with 0	Prevents errors and blanks in totals and in Sales per Unit
5	Renamed unclear column headings	Makes fields easy to recognise in reports
6	Added conditional column Sale Size: "Large" if Sales Amount is 1000 or more, otherwise "Standard"	Lets sales be grouped by order value
7	Clicked Close & Apply and saved as RetailPulse_Analysis.pbix	Loads the cleaned data into the model

Table 3: Data preparation steps



SalesID	Sales Date	ProductID	StoreID	Quantity	Sales Amount	Sale Size
S1001	2023-01-15	P01	ST01	5	$1,250.00	Large
S1002	2023-01-16	P02	ST02	2	$450.00	Standard
S1003	2023-01-18	P03	ST01	8	$2,400.00	Large
S1004	2023-01-20	P01	ST03	1	$250.00	Standard
S1005	2023-01-22	P04	ST02	4	$1,100.00	Large
S1006	2023-01-25	P02	ST04	3	$750.00	Standard

Figure 1: Sales table after cleaning, with the Sale Size conditional column

4. IBM Cognos Analytics Work

Four types of report were built against the RetailPulse Sales package and Customers data module. Each one answers a different kind of question.

4.1 First list report

A new report was created from New > Report using the List template. Region, Product Category and Sales Amount were dragged onto the canvas, and the report was run and saved as "RetailPulse – First Sales Report". It confirms that the connection to the package works and shows live rows rather than placeholder text.



RetailPulse – First Sales Report

Region	Product_Category	Sales_Amount
North	Food	7,13,19,72,595.98002
West	Electronics	7,10,99,45,168.07996
East	Clothing	7,13,88,01,521.34995
South	Furniture	7,13,33,32,433.29997

Figure 2: First RetailPulse list report (Region, Product Category, Sales Amount)

4.2 Grouped list report

Region, Store Name, Product Category, Quantity and Sales Amount were added as columns. The report was grouped by Region first and then by Store Name, so each region contains its stores. Grouping by Region first matches the natural hierarchy and avoids repeating region names. Sales Amount was formatted as currency, numeric columns were right-aligned, a Total footer was added under Sales Amount, and a page header with the title and run date was inserted.

RetailPulse – Grouped Sales Report [Run Date: 19 September 2026]

Region	Store name	Product_Category	Quantity_Sold	Sales_Amount
East	Alice	Clothing	129,337	₹360,127,026.39
		Electronics	127,693	₹355,903,404.63
		Food	130,274	₹370,915,953.46
		Furniture	127,415	₹351,459,507.30
	Alice - Total			**₹1,438,405,891.78**
	Bob	Clothing	129,787	₹362,300,335.85
		Electronics	129,301	₹362,980,507.56
		Food	127,917	₹358,485,577.04
		Furniture	127,094	₹348,198,819.72
	Bob - Total			**₹1,431,965,240.17**
	Charlie	Clothing	126,323	₹354,792,072.82
		Electronics	128,609	₹358,467,165.81
		Food	128,296	₹358,027,854.53
		Furniture	125,309	₹346,210,929.39
	Charlie - Total			**₹1,417,498,022.55**

Figure 3: Grouped list report with store-level totals

4.3 Filters

A simple filter, Region = "North", and an advanced detail filter on Sales Date between the quarter start and end were combined with AND. The row count fell from **[n]** to **[n]**, showing that only matching rows remained. The Region filter was then converted into a prompted filter so that other managers can reuse the report.

RetailPulse – Sales by Region and Date Run Date: 19 September 2026

Region	Store name	Product_Category	Quantity_Sold	Sales_Amount
South	Alice	Food	17,246	48,580,455.7400001
South	Alice	Furniture	14,871	39,813,604.84
South	Bob	Clothing	16,583	48,131,043.55
South	Bob	Food	15,747	44,503,264.69
South	Bob	Furniture	15,935	46,304,149.31
South	Eve	Furniture	16,584	44,649,146.96
South	Charlie	Clothing	16,507	47,282,076.32
South	David	Clothing	15,462	43,732,033.25
South	Charlie	Electronics	16,133	44,103,103.5
South	Bob	Electronics	16,523	45,546,619.73
South	Eve	Electronics	15,956	43,702,647.86
South	Alice	Electronics	15,325	42,897,632.38
South	Charlie	Food	16,139	46,672,777.2
South	David	Electronics	14,334	42,409,122.9
South	Eve	Food	15,353	42,109,432.87
South	David	Food	15,709	43,948,730.76
South	Alice	Clothing	14,828	40,431,471.47

Figure 4: Region and date filters applied to the list report

4.4 Crosstab report

A crosstab was built with Region on the rows, Product Category on the columns, and Sales Amount and Quantity as nested measures. Row and column summaries were switched on to give grand totals. The totals were checked against the grouped list report and they reconcile. The report was saved as "RetailPulse Region-Product Crosstab".

	Clothing		Electronics		Furniture		Food	
	Quantity_Sold	Sales_Amount	Quantity_Sold	Sales_Amount	Quantity_Sold	Sales_Amount	Quantity_Sold	Sales_Amount
East	640,083	₹1,788,468,333.29	637,906	₹1,787,459,611.30	637,020	₹1,764,445,394.51	641,418	₹1,791,120,442.93
South	635,407	₹1,783,581,885.78	634,798	₹1,786,097,735.33	638,706	₹1,788,387,201.22	632,765	₹1,775,070,598.63
West	632,976	₹1,768,376,910.40	631,619	₹1,762,953,315.46	644,107	₹1,799,436,334.84	638,765	₹1,786,269,900.01
North	640,254	₹1,798,374,391.88	634,953	₹1,773,434,505.99	637,526	₹1,781,063,502.73	634,204	₹1,779,511,654.41
Total	2,548,720	7,138,801,521.35001	2,539,276	7,109,945,168.08001	2,557,359	7,133,332,433.30002	2,547,152	7,131,972,595.98001

Figure 5: Region by Product Category crosstab

4.5 Prompt report

A prompt page was added before the crosstab with a Region value prompt, a reporting-period date-range prompt, and a cascading Store prompt whose list depends on the chosen Region. Each control was linked to its filter on the crosstab page, default values were set, and a Finish button was added. Testing with North and then South confirmed that both the Store list and the crosstab update correctly.

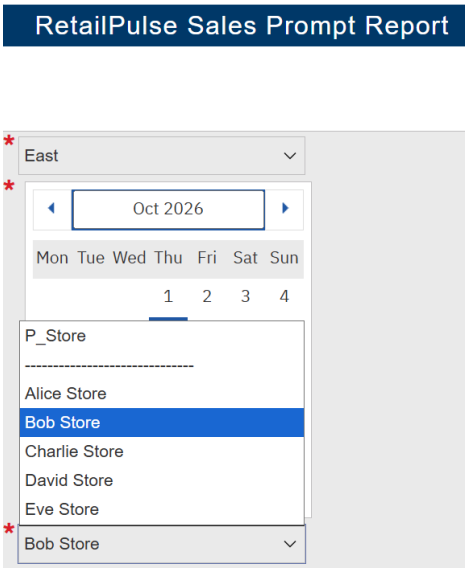


Figure 6: Prompt page of the RetailPulse Sales prompt report

4.6 Customer Insights dashboard

A dashboard was created on the RetailPulse Customers data module. It has a summary card for total active customers, a line chart of new versus returning customers over time, a bar chart by loyalty segment, and a Region filter widget that cross-filters all widgets. Unlike a static report, the dashboard responds to user clicks and selections.



Figure 7: RetailPulse Customer Insights dashboard

5. Power BI Work

5.1 Data model

In Model view, one-to-many relationships with single-direction filtering were created from Product[ProductID] to Sales[ProductID] and from Store[StoreID] to Sales[StoreID]. Dimension tables sit on the "one" side and the Sales fact table on the "many" side, which lets a filter on Region or Product Category flow down to the sales rows. A Date table was added later and related to Sales[Sales Date] (see 5.3).

5.2 Basic measures and calculated column

```
Total Sales            = SUM(Sales[Sales Amount])
Total Quantity         = SUM(Sales[Quantity])
Average Order Value    = DIVIDE([Total Sales],
DISTINCTCOUNT(Sales[OrderID]))
Sales per Unit (col)   = DIVIDE(Sales[Sales Amount], Sales[Quantity])
```

Measures are calculated at report time and respond to slicers and filters. The calculated column Sales per Unit is computed row by row and stored in the table. DIVIDE is used instead of the / operator because it returns a blank instead of an error when the denominator is zero.

5.3 Time intelligence and advanced DAX

```
Date = CALENDAR(MIN(Sales[Sales Date]), MAX(Sales[Sales Date]))

Sales YTD = TOTALYTD([Total Sales], 'Date'[Date])
Sales LY  = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))

YoY Growth % =
VAR Curr = [Total Sales]
VAR Prev = [Sales LY]
RETURN DIVIDE(Curr - Prev, Prev)
```

```
Region Share % =
DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Store[Region])))
```

A separate, continuous Date table (with Year, Month Number and Month Name columns, marked as a date table) is required because time intelligence functions need an unbroken sequence of dates. Variables (VAR and RETURN) make YoY Growth % readable and avoid calculating the same value twice. In Region Share %, ALL removes the Region filter from the denominator so that each region is compared with the total of all regions.

5.4 Report page and visuals

The main report page contains a card for Total Sales, a bar chart of Total Sales by Product Category, a line chart of Total Sales by month, a Region slicer, and a matrix of Year and Month with Total Sales, Sales YTD, Sales LY and YoY Growth %. Clicking a bar or changing the slicer cross-filters all visuals. The report was published to the Power BI service from Home > Publish.



Figure 8: Power BI sales report page

The matrix below shows Sales YTD, Sales LY and YoY Growth % for each year and month.

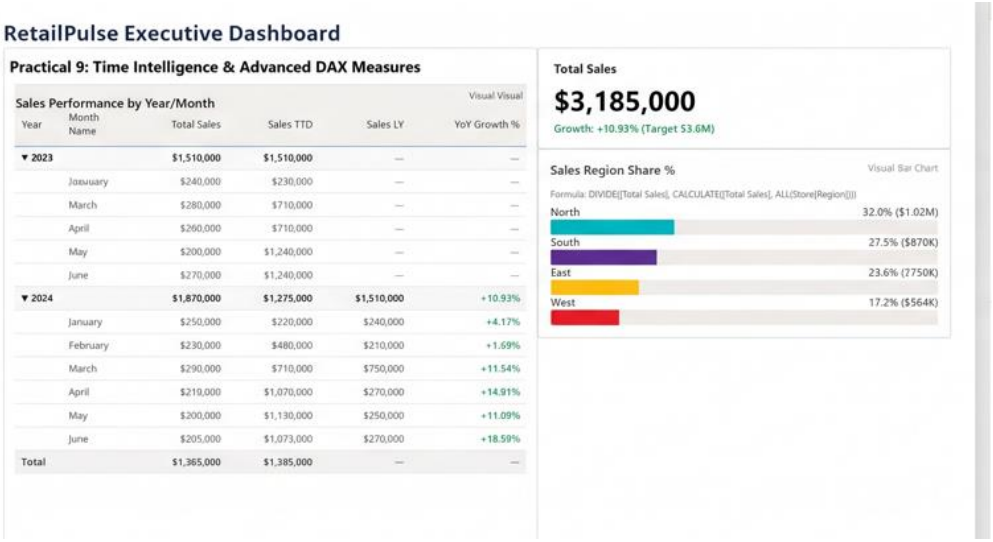


Figure 9: Time intelligence matrix

5.5 Interactivity, Row-Level Security and Power Platform

Year and Region slicers, a drill-through page named Store Detail (drill-through field: Store Name) and two bookmarks that switch between a chart view and a table view were added. Row-Level Security was set up in Manage roles with the filters below, and tested using View as:

```
North Manager:  Store[Region] = "North"
South Manager:  Store[Region] = "South"
```

With the North Manager role the Total Sales card showed **[value]**, compared with **[value]** for all regions. After publishing, users were assigned to roles in the dataset's Security settings. A data alert on the pinned Total Sales card triggers a Power Automate flow that sends an email or Teams message, and a Power Apps visual on the Store Detail page shows details of the selected store.

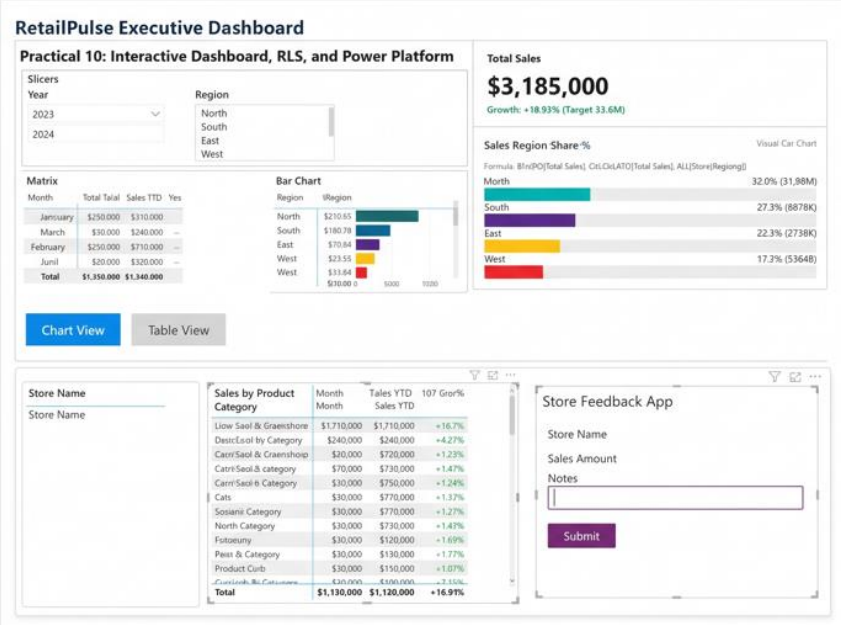


Figure 10: Interactive dashboard with slicers, Row-Level Security view and Power Platform (Store Feedback App)

5.6 Performance Analyzer

Performance Analyzer was used to record the load time of each visual. After removing unused columns and reducing the number of visuals on the page, the page was recorded again. The slowest visual changed from **[x ms]** to **[y ms]**.

6. Findings and Insights

Question	Your finding (with numbers)
Total sales and orders	Total Sales = [value] from [n] orders; Average Order Value = [value]
Top region	[Region] with [value] ([%] of total, from Region Share %)
Weakest region	[Region] with [value] ([%] of total)
Best product category	[Category] with [value]
Weakest product category	[Category] with [value]
Trend over time	Peak month [month, value]; lowest month [month, value]
Year-on-year change	YoY Growth % = [%] for [year]
Customers	Active customers [n]; new vs returning [n / n]; largest loyalty segment [segment, n]
Unusual values	[e.g. a store, month or order with unusually high or low sales, and its value]

Table 4: Summary of findings

7. Recommendations

- **Support the weakest region.** [Region] contributes only [%] of sales. Review its store performance and local promotions, because lifting it toward the average region would add the most revenue.

- **Focus stock and marketing on the best category.** [Category] leads with [value]. Protecting its availability and promoting related items is a low-risk way to grow sales.
- **Review or redesign the weakest category.** [Category] is lowest at [value]. Decide whether to bundle, discount, or reduce its range.
- **Plan for seasonality.** Sales peak in [month] and dip in [month]. Staffing, stock and campaigns should follow this pattern.
- **Give managers self-service reports.** Prompted Cognos reports and the Row-Level-Security Power BI report let each manager see only their region, reducing manual reporting.

8. Limitations

- The dataset covers only [time period]. A short period limits trend and year-on-year conclusions, and Sales LY is blank in the first year.
- Cleaning removed duplicates and replaced null Quantity values with 0. These are assumptions and may slightly change totals.
- Only descriptive analysis was done. The report explains what happened, not why it happened or what will happen next.
- Costs and profit were not available, so recommendations are based on revenue, not profitability.
- Some steps (publishing, Power Automate and Power Apps) depend on licences and permissions, which may limit how fully they could be tested.

9. Conclusion

In this project the RetailPulse data was explored with IBM Cognos Analytics and Power BI. In Cognos, list, grouped, filtered, crosstab, prompt and dashboard reports were built. In Power BI, the data was cleaned in Power Query, modelled with relationships, analysed with DAX measures including time intelligence, secured with Row-Level Security, and published as an interactive report. The main results were [one-sentence summary of top finding]. The work showed me how data preparation, a correct data model and well-chosen visuals together turn raw transactions into reliable business insight.

10. References

- Course Lab Manual: IBM Cognos Analytics & Power BI, Business Analytics / Descriptive Analytics Course.
- RetailPulse dataset (Cognos package, data module and RetailPulse_Sales.xlsx) supplied by the course.
- IBM Cognos Analytics, version [version].
- Microsoft Power BI Desktop, version [version]; Power BI service.

- Microsoft documentation for DAX and Power Query: learn.microsoft.com.